MATHESHWARAN N

Madurai/Chennai, Tamilnadu, India • +91 8608391949 • matheshwaran1300@gmail.com
Portfolio: mathesh.pages.dev | GitHub: github.com/mathesh-cpu | Linkedin: linkedin.com/in/mathesh-waran-n-

PROFESSIONAL SUMMARY

Eager and detail-oriented Entry-Level Cloud & Systems Engineer with hands-on lab experience across Linux, Docker, AWS, and Google Cloud environments. Practical foundation in deploying containerized web apps, writing Bash scripts for automated tasks, managing basic PostgreSQL databases, and setting up system monitoring using Prometheus and Grafana. Passionate about learning production operations, troubleshooting Linux systems, and supporting cloud infrastructure teams.

TECHNICAL SKILLS

- **Cloud Platforms:** Amazon Web Services (EC2, S3,IAM Roles & Policies), Google Cloud Platform (Foundational labs via Cloud Skills Boost)
- **Linux & Automation:** Ubuntu Linux, Bash Shell Scripting, Scheduled Cron Jobs, System Navigation & Permissions
- **Containers & Orchestration:** Docker, Docker Compose (Multi-Container Docker Compose)
- **Databases:** PostgreSQL (Containerized deployment, Persistent Volumes, backups with pg_dump)
- **Monitoring & Tools:** Prometheus, Grafana, Git, GitHub

HANDS-ON PROJECTS

Resilient Cloud Architecture & Backup Lab | AWS, Linux, Docker, PostgreSQL

GitHub: github.com/mathesh-cpu/aws-resilient-cloud-architecture

- **Isolated Container Deployment:** Configured a multi-tier web application and PostgreSQL database using Docker Compose, implementing dedicated volume mounts to ensure persistent data retention across container lifecycle restarts.
- **Linux Memory Optimization:** Prevented Out-Of-Memory (OOM) database process terminations on a constrained EC2 instance by allocating a 2GB swap partition, stabilizing the container under write traffic spikes.
- **Least-Privilege Cloud Security:** Secured EC2-to-S3 backup access via an attached AWS IAM Instance Profile, eliminating the security risk of storing hardcoded API credentials or secret keys on the filesystem.
- **Automated Off-Site Backups:** Engineered a scheduled cron Bash pipeline generating automated logical PostgreSQL snapshots (pg_dump), streaming compressed archives to Amazon S3 to protect against local disk corruption.
- **Disaster Recovery Validation:** Verified end-to-end data recoverability by simulating a database drop scenario and successfully executing a complete restoration from remote S3 snapshots.
- **Infrastructure Telemetry:** Integrated Prometheus and Node Exporter with a Grafana dashboard, providing immediate visual tracking of host CPU, memory pressure, and container status.

EDUCATION

Master of Science (M.Sc.) in Data Science | 2024 – 2026
Hindustan Institute of Technology and Science, Chennai

Bachelor of Vocation (B.Voc.) in Data Science | 2021 – 2024
The American College, Madurai

CERTIFICATIONS & LEARNING PATHS

Associate Cloud Engineer Learning Path & Hands-on Labs — Google Cloud Skills Boost